

Expense Tracking & Budget Control System

Multi-Tenant SaaS Platform — OOP Training Program

Any organisation can sign up, invite their team, set budgets, and track expenses — each in their own private workspace.

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1. Purpose

Companies need a simple way to track employee expenses, set yearly spending budgets for each expense category, and get alerts before they overspend. This system serves **multiple organisations at once** — each organisation gets its own private workspace where only their team can see their data.

- Each organisation signs up and chooses a subscription plan.
- The organisation admin invites team members and assigns them roles.
- Team members record their expenses under named categories.
- The system monitors spending against the budget and sends alerts.
- Managers and Admins can view the full team's monthly spending report.

What Makes It Multi-Tenant?

A single running application serves many organisations at the same time. Each organisation's expense data, categories, budgets, and users are completely private. One company cannot see another company's data — even though they are all using the same system.

Example: TechCorp BD and Medicore Ltd both use the same Expense Tracker. TechCorp's employees can only see TechCorp's expenses. Medicore's team sees only their own. The system keeps them separate automatically.

2. Business Scenario

TechCorp BD is a software company with 8 employees. The company owner, Rahim Ahmed, signs up for the platform, chooses the **Pro Plan**, and creates the company workspace. He sets up five expense categories and a yearly budget for each. He then invites his team — some as regular Members who add their own expenses, and one senior as a Manager who can review the team's full spending.

Every time a team member spends money — a taxi ride, a team lunch, a software subscription — they open the app and record the expense. The system checks if the spending is close to or over the budget, and alerts the Admin and Manager if needed. At the end of each month, the Manager opens the Monthly Summary to see how much each category has been spent, and whether the team is on track.

Daily Operations

Who	What They Do Daily
Admin Rahim Ahmed	Monitors budget health on the dashboard, manages team members, handles billing
Manager Nusrat Jahan	Reviews team expenses, approves over-budget entries, views monthly report
Member Fahmida, Tarek, Karim...	Adds their own expenses; can only see their own history

Expense Categories (TechCorp BD)

Category	Monthly Budget	Limit Type
Travel & Transport	₹30,000	Hard Limit
Meals & Entertainment	₹25,000	Soft Limit
Software Subscriptions	₹40,000	Soft Limit
Office Supplies	₹20,000	Soft Limit
Training & Courses	₹35,000	Soft Limit

Soft Limit vs Hard Limit

Limit Type	What Happens When Budget Is Exceeded
Soft Limit	The expense is saved. The system marks it as "Over Budget" and sends an alert. No approval needed.
Hard Limit	The expense is held. The member cannot complete the entry until a Manager or Admin approves it.

3. Users & Roles

The system has four types of users. Each organisation has its own set of users, separate from all other organisations.

Role	Who They Are	What They Can Do	What They Cannot Do
Super Admin	The platform operator (not part of any organisation)	View all organisations, manage subscription plans, monitor platform health	Cannot see any organisation's expense data
Tenant Admin	The organisation owner or IT lead	Invite & remove users, set roles, create categories, set budgets, view all expenses, manage billing	Cannot access other organisations' workspaces
Manager	Team lead or finance officer	View all team expenses, approve over-budget entries, view monthly report, add own expenses	Cannot manage users or change billing plan
Member	Regular employee	Add, edit, and view their own expenses only	Cannot see other members' expenses or reports

Note: One person can belong to multiple organisations with different roles in each. For example, a consultant might be a Manager in TechCorp BD and a Member in another company — using the same login.

4. Subscription Plans

Each organisation chooses a plan when they sign up. The plan determines how many users they can invite, how many expense categories they can create, and how far back they can see their history.

Free Starter \$0 forever Up to 3 users Up to 10 categories 6 months expense history Basic monthly report In-app notifications only	Most Popular Pro \$49 / month Up to 15 users Unlimited categories Full expense history Advanced reports & member filter Email & SMS alerts Manager approval workflow	Enterprise Enterprise Custom pricing Unlimited users Unlimited categories Full history + archive export Custom reports & data export Single sign-on (SSO) Priority support
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Plan Limits Enforced by the System

- If an organisation is on the Starter plan and already has 3 users, inviting a 4th user is blocked — the system shows an upgrade prompt.
- If a Starter organisation tries to create an 11th expense category, the system blocks it and suggests upgrading.
- Starter organisations cannot see expense history older than 6 months. The data is kept, just hidden until they upgrade.
- An organisation cannot downgrade to a lower plan if their current usage exceeds that plan's limits.

5. Core Features

 Organisation Workspace Each organisation gets a fully private workspace. Sign up, choose a plan, and your team's data is isolated from everyone else.	 Team Invitations Admins invite team members by email and assign them a role. Invitation links expire after 48 hours. Accepted members can log in immediately.	 Expense Categories Create named categories to organise expenses (Travel, Meals, Software, etc.). Categories can be deactivated without losing old expense history.
 Add Expense Members record daily expenses with date, category, amount, and a short description. The system warns if the same expense looks like a duplicate.	 Yearly Budget Setup Admin sets a budget for each expense category for the year. Only the current year's budget can be changed. Past year budgets are locked.	 Budget Monitoring The system checks every new expense against the category's budget. Alerts go out at 80% usage. Soft or Hard Limit rules apply at 100%.



Manager Approval

When a Hard Limit budget is exceeded, the entry waits for Manager or Admin approval. The Manager can approve or reject with a note.



Monthly Summary

Managers and Admins see a full monthly report: total spent, budget vs actual per category, and which members spent what.



Notifications

In-app alerts for budget thresholds, over-budget entries, approval requests, and plan events. Each alert fires only once per period — no spam.



Adjustments & Corrections

If an expense was entered by mistake, the correction is recorded as a separate Adjustment Entry — the original record is never changed.

MVP Features (Build These First)

- Organisation sign-up and workspace creation
- User invitation with role assignment
- Login with organisation context
- Create and manage expense categories
- Add and view expenses (members see only their own)
- Set yearly budget per category (Soft Limit only)
- In-app budget alert at 80% threshold
- Monthly summary screen (Managers and Admins only)
- Platform dashboard for Super Admin (organisation list)

6. Main Screens — UI Mocks

6.1 Login Screen

Entry Point

Log in to Your Workspace

Email Address

rahim@techcorp.com.bd

Password

Enter your password

Log In

Forgot Password?

Example accounts: rahim@techcorp.com.bd (Admin) | nusrat@techcorp.com.bd (Manager) | fahmida@techcorp.com.bd (Member)

New Organisation? Sign Up

Organisation Name

TechCorp BD

Your Work Email

owner@yourcompany.com

Create Workspace →

6.2 Organisation Onboarding

New Signup

✓ Create Account

2 · Set Up Organisation

3 · Choose Plan

4 · Invite Team

Step 2 — Set Up Your Organisation

Organisation Name *

TechCorp BD

Industry

Technology

Team Size

6-15

Currency

BDT (b)

Continue →

Step 3 — Choose a Plan

Starter

Pro Popular

Enterprise

Price	Free	\$49 / month	Contact us
Users	3	15	Unlimited
Categories	10	Unlimited	Unlimited
History	6 months	Full	Full + Archive
Select Select Pro Contact Sales			

6.3 Dashboard

Admin View

TechCorp BD — June 2026 Overview

Team Spent

₹86,400

Total Budget

₹1,50,000

Team Members

8 / 15

Pending Approvals

2

Budget Health — All Categories

Travel & Transport

₹32,100 / ₹30,000 — Over Budget

Meals & Entertainment

₹20,400 / ₹25,000 — 82%

Software Subscriptions

₹18,000 / ₹40,000 — 45%

Office Supplies

₹9,400 / ₹20,000 — 47%

Training & Courses

₹6,500 / ₹35,000 — 19%

Recent Expenses (All Team)

Date	Member	Category	Description	Amount	Status
15 Jun	Fahmida	Travel	Taxi to client — Gulshan	₹1,200	Pending Approval
12 Jun	Tarek	Meals	Team lunch	₹3,800	Saved
10 Jun	Karim	Software	GitHub Pro	₹1,500	Saved

Add Expense

Monthly Summary

Manage Team

6.4 Add Expense Screen

Member View

Add New Expense — Fahmida Akter · TechCorp BD

Date *

15/06/2026

Expense Category *

Travel & Transport

Amount (₹) *

1200

Description

Taxi to client site — Gulshan

⚠ **Budget Impact:** Travel & Transport is already over its ₹30,000 budget (currently ₹32,100). This category has a **Hard Limit** — this entry will need Manager or Admin approval before it is saved.

Submit for Approval

Cancel

ⓘ As a **Member**, you can only view and edit your own expenses. Your Manager can see all team expenses.

6.5 Manage Team Screen

Admin Only

Invite a New Team Member

Email Address *

Role *

karim@techcorp.com.bd

Manager

Send Invite

Plan: Pro — 8 of 15 seats used. You can invite 7 more members.

Current Team — TechCorp BD

Name	Email	Role	Status	Action
Rahim Ahmed	rahim@techcorp.com.bd	Admin	Active	—
Nusrat Jahan	nusrat@techcorp.com.bd	Manager	Active	Change Role
Fahmida Akter	fahmida@techcorp.com.bd	Member	Active	Remove
Karim Hasan	karim@techcorp.com.bd	Member	Invited	Resend

6.6 Budget Setup Screen

Admin Only

Set Yearly Budget — 2026

One budget per category per year. Budgets for previous years are locked and cannot be changed. Categories without a budget are treated as Unlimited — no alerts will fire for them.

Category	Budget Amount (b)	Limit Type	Status
Travel & Transport	30000	Hard Limit	Set
Meals & Entertainment	25000	Soft Limit	Set
Software Subscriptions	40000	Soft Limit	Set
Office Supplies	20000	Soft Limit	Set
Entertainment	Not set	—	Unlimited

Save Budget

Reset Changes

6.7 Monthly Summary Screen

Manager & Admin

June 2026 — All Team

All Members

June 2026

Total Spent

₹86,400

Total Budget

₹1,50,000

Remaining

₹63,600

Over Budget

1 category

Category	Budget	Spent	Remaining	Status
Travel & Transport	₹30,000	₹32,100	-₹2,100	Over Budget
Meals & Entertainment	₹25,000	₹20,400	₹4,600	82% Used
Software Subscriptions	₹40,000	₹18,000	₹22,000	On Track
Office Supplies	₹20,000	₹9,400	₹10,600	On Track
Training & Courses	₹35,000	₹6,500	₹28,500	On Track

Export Report

View by Member

6.8 Notifications Screen

All Roles

Recent Alerts — TechCorp BD

Approval Required — Travel & Transport

Fahmida Akter submitted ₹1,200 (Taxi — Gulshan). Travel is a Hard Limit category and is already over budget. Manager or Admin must approve.

15 Jun 2026 - 03:14 PM - Review & Approve

Travel & Transport — Over Budget

Team spending b32,100 exceeds the b30,000 Hard Limit budget (107%). Two entries are now waiting for approval.

14 Jun 2026 · 11:42 AM

Meals & Entertainment — 80% Alert

Team has spent b20,400 of the b25,000 budget (82%). This alert will not repeat until next month.

10 Jun 2026 · 09:15 AM

New Member Joined

Fahmida Akter accepted the invitation and joined TechCorp BD as a Member.

10 Jun 2026 · 08:50 AM

Team Seats — 8 of 15 Used

Your Pro plan allows 15 users. You have 8 active members. 7 seats remaining before you need to upgrade.

01 Jun 2026 · 08:00 AM

Mark All as Read

7. System Flows

Flow 1: Organisation Sign Up

- Owner opens the website and clicks "Create Workspace".
- Owner enters organisation name, email, and password.
- System creates the organisation's private workspace.
- Owner chooses a subscription plan (Starter / Pro / Enterprise).
- Owner is automatically set as the Tenant Admin.
- Onboarding wizard guides the owner: Create categories → Set budgets → Invite team.
- Workspace is ready. Team members receive invitation emails.

Flow 2: Add an Expense (Member)

- Member logs in and opens the Add Expense screen.
- Member selects date, category, enters amount and a short description.
- System checks: amount must be positive; date cannot be in the future.
- System checks for a duplicate entry (same date, amount, category, and description).
- If duplicate found: system shows a warning. Member can still choose to save.
- System checks the category's budget.
- 7a. Soft Limit:** Expense is saved. If over budget, it is marked "Over Budget" and an alert is sent.
- 7b. Hard Limit exceeded:** Expense is held as "Pending Approval". Manager/Admin is notified.
- Member sees a confirmation or a "Waiting for Approval" message.

Flow 3: Over-Budget Approval (Manager)

- Manager opens the Notifications screen and sees an approval request.
- Manager clicks "Review & Approve" to see the expense details and current budget status.
- 3a. If approved:** Manager clicks Approve (optionally adds a note). Expense is saved. Member is notified.
- 3b. If rejected:** Manager clicks Reject and enters a reason. Expense is discarded. Member is notified with the reason.

Flow 4: Invite a Team Member (Admin)

- Admin opens the Manage Team screen.
- System checks if the team has reached the plan's user limit.
- If limit reached: system shows an upgrade prompt. Invitation is blocked.
- Admin enters the new member's email and selects their role.
- System sends an invitation email. The link expires in 48 hours.
- Member clicks the link, sets a password, and joins the workspace.
- Admin sees the member's status change from "Invited" to "Active".

Cancellation / Correction Flow

1. Member or Admin identifies an expense that was entered incorrectly (e.g. a refund was received).
2. They open the Adjustment screen and link to the original expense.
3. They enter the correction amount and a reason.
4. System saves the Adjustment Entry. The original expense record is never changed.
5. The monthly report reflects the adjustment correctly. Negative amounts never appear anywhere.



8. Important Business Rules

Expense Entry Rules

- An expense amount must always be positive. Zero or negative amounts cannot be saved.
- An expense date cannot be set to a future date.
- Every expense must have a category. A category cannot be left blank.
- If the same date, amount, category, and description already exist, the system warns the user — but does not block them.
- A member can only edit or delete their own expenses, and only if they belong to the current year. Previous-year expenses are locked.
- Corrections and refunds must be entered as Adjustment Entries. The original expense is never modified.

Budget Rules

- Only one budget can be set per category per year, per organisation.
- If no budget is set for a category, spending is treated as Unlimited — no alerts will fire.
- Only the Tenant Admin can set or change a budget. Only the current year's budget can be changed.
- Budget amounts can include decimals (e.g. £15,000.50).



Expense Category Rules

- Category names must be unique within the same organisation (not case-sensitive — "Travel" and "travel" count as the same).
- A category name cannot be blank or contain only spaces. Maximum 50 characters.
- Only the Admin can create, edit, or deactivate categories.
- Deactivating a category hides it from the expense entry screen, but all past expenses recorded under it remain visible in reports.
- A deactivated category can be reactivated at any time.
- Starter plan organisations cannot create more than 10 categories. Creating an 11th is blocked until they upgrade.

Overspending Rules

- When spending reaches 80% of the budget, the system sends an alert. This alert fires only once per month per category.
- **Soft Limit:** The expense is saved and marked "Over Budget". An alert is sent to Admins and Managers.
- **Hard Limit:** The expense cannot be saved without Manager or Admin approval. The entry waits in "Pending" state.
- An approved Hard Limit entry is saved with a note of who approved it and when.
- A rejected Hard Limit entry is permanently discarded. The member is told the reason.



Organisation & Workspace Rules

- Each organisation's data is completely private. One organisation cannot see another's expenses, categories, or budgets.
- Only the Tenant Admin can invite users, change roles, or remove members.
- When a member is removed from an organisation, their past expense entries remain in the organisation's records.
- A member can belong to multiple organisations and will see a workspace selector after logging in.
- Invitation links expire after 48 hours. If not accepted, the Admin must resend.

Notification Rules

- In-app notifications are on by default for all users.
- Email and SMS alerts are available on Pro and Enterprise plans only.
- Notifications for the same event (e.g. 80% alert for the same category) are sent only once per period to avoid spam.
- Approval request notifications are sent immediately to all Managers and Admins in the organisation.



9. Example Report Data

Monthly Summary Report — TechCorp BD, June 2026

Category	Budget (£)	Spent (£)	Remaining (£)	Status
Travel & Transport	30,000	32,100	-2,100	Over Budget
Meals & Entertainment	25,000	20,400	4,600	82%
Software Subscriptions	40,000	18,000	22,000	On Track
Office Supplies	20,000	9,400	10,600	On Track



Training & Courses	35,000	6,500	28,500	On Track
TOTAL	1,50,000	86,400	63,600	

Member-Wise Spending — June 2026 (Manager View)

Member	Role	Total Expenses	Number of Entries
Rahim Ahmed	Admin	₹12,400	8
Nusrat Jahan	Manager	₹18,200	11
Fahmida Akter	Member	₹22,100	14
Tarek Mahmud	Member	₹19,800	12
Others (4 members)		₹13,900	22
TOTAL		₹86,400	67 entries

Super Admin — Platform Overview

Organisation	Plan	Users	Status	Joined
TechCorp BD	Pro	8 / 15	Active	01 Jun 2026
Medicore Ltd	Starter	3 / 3	Plan Limit Reached	14 May 2026
NovaSoft Inc	Enterprise	47 / ∞	Active	10 Apr 2026
GreenLeaf Co	Starter	2 / 3	Active	20 Jun 2026

10. Technical Specification

You can use any programming language you want.

You can use any database — or even an in-memory data store for the first version.

The project can be a console, desktop, or web application.

Focus on the object design: how are Organisations, Users, Expense Categories, Expenses, and Budgets represented as classes, and how do the business rules connect them?

Suggested build order: Start with a single organisation (no multi-tenancy yet). Get expense entry, category management, and budget alerts working first. Once that works, add a second organisation and enforce data separation between the two. Add subscription plan limits last.

[See the main program](#)

11. Advanced Topics — For Discussion

The three questions below are not part of the MVP. They are discussion topics for when the system grows into a large-scale SaaS product serving thousands of organisations and millions of daily expense entries. Think about them from a business and product perspective — not just a technical one.

US-20 — How to Make the Daily Expense Report Load Faster?

The Problem: When TechCorp BD has 8 employees adding a few expenses each day, the Monthly Summary loads in under a second. But when the platform grows to 10,000 organisations — each with hundreds of employees adding expenses every day — that same report could take over a **minute** to load. That is not acceptable for a SaaS product. The goal is to bring it down to **3 to 5 seconds**.

Why does it get slow? Every time someone opens the Monthly Summary, the system currently reads every single expense entry for that month and adds them all up on the spot. With a small team this is fine. With millions of entries across thousands of organisations, it becomes very slow.

Discussion: What Can Be Done?

Idea	How It Helps (in plain terms)
Pre-calculate daily totals in the background	Instead of adding up all expenses every time a report is opened, the system quietly runs a background job at the end of each day and stores the day's total per category. When someone opens the report, it reads the pre-stored totals instead of scanning millions of rows from scratch. <i>Think of it like a bank balance: your bank does not add up every transaction from day one every time you check your balance — it keeps a running total and just updates it.</i>
Save a snapshot of each month's report	Once a month has ended, its report numbers will never change. The system can save a permanent copy of the finished monthly report. Future requests for that month just read the saved copy — no calculations needed at all.

Cache recent reports temporarily

If 50 people in the same organisation all open the June report in the same hour, the system should not recalculate it 50 times. It calculates once, saves the result briefly, and serves the saved result to everyone else who asks for it in the same window.

Discussion question: If you pre-calculate and store daily totals, what happens when a Manager approves a pending expense that was submitted three days ago? How does the stored total get corrected?

US-21 — How to Use AI/ML for Smarter Duplicate Detection?

The Problem: The current system only catches exact duplicates — the same date, amount, category, and description must all match. But in real life, the same expense can be entered twice with slightly different wording. A human can spot this instantly; the current system cannot.

Examples the Current System Misses

First Entry	Second Entry	Are They the Same?
Travel · ₹1,200 · "Taxi to Gulshan"	Travel · ₹1,200 · "Cab to client office Gulshan"	Probably yes
Meals · ₹3,800 · "Team lunch"	Meals · ₹3,800 · "Lunch — team meeting"	Probably yes
Software · ₹1,500 · "GitHub Pro subscription"	Software · ₹1,500 · "GitHub monthly fee"	Very likely yes
Travel · ₹1,200 · "Taxi to Gulshan"	Travel · ₹1,200 · "Taxi to Mirpur"	Probably no — different destination

How Could AI Help?

An AI system can be trained to read the description of a new expense and compare it to recent entries for the same user. If the descriptions are similar in meaning — even if the exact words are different — the AI can flag it as a possible duplicate and ask the member to confirm before saving.

The AI does not block the entry. It suggests: *"This looks similar to an entry you saved on 14 June. Are you sure this is a different expense?"* — and the member can confirm or cancel.

Discussion question: The AI will sometimes be wrong — it might flag two different expenses as duplicates when they are not. What is the cost of a false warning vs. a missed duplicate? How confident should the AI need to be before it shows a warning?

US-22 — AI-Based Chatbot for Report Queries

The Idea: Instead of navigating through multiple screens to find a number, a Manager or Admin can simply type a question in plain language and get an instant answer — without opening a single report screen.

Example Questions the Chatbot Could Answer

Question (typed by user)	What the chatbot finds and shows
"What did we spend the most on last quarter?"	Travel & Transport — ₹98,400 (Q2 2026)
"Which team member has the highest expenses this month?"	Fahmida Akter — ₹22,100 across 14 entries in June 2026
"Are we going to go over our Meals budget this month?"	At current pace, you will reach ₹26,400 against a ₹25,000 budget by month end.
"Show me all travel expenses above ₹2,000 in May."	Lists the matching entries directly in the chat
"How does this month compare to last month?"	June ₹86,400 vs May ₹74,200 — 16% higher this month. Biggest increase: Travel.

Who Benefits and How

Role	How the Chatbot Helps Them
Admin	Get a quick budget health check without opening the report screen. Ask "are we on track?" at any time.
Manager	Compare member spending, spot patterns, and answer questions from team members without manually digging through data.
Member	Ask "how much budget is left in Travel this month?" and get an instant answer — scoped only to their own data.

Important rule: The chatbot must respect the same access rules as the rest of the system. A Member asking "what did the whole team spend?" must get the same response as if they had tried to open the Monthly Summary screen — access denied.

Discussion question: The chatbot only has access to this organisation's data. How do you make sure it cannot accidentally answer a question using another organisation's numbers?

